a AVISA instruction format (64 bit)



opcode	operation	issue cost
0x01	FMA.TILE	1 lane-cycle
0x08	STENCIL3D (7-point, fused)	1 lane-cycle / point
0x10	ADVECT.SL (8-point gather)	2 lane-cycles
0x14	ENS.SPREAD (reduce 256 lanes)	9 cycles
0x18	ENS.TL (twin trajectory step)	2 lane-cycles

b Arithmetic formats

format	mantissa	lanes	peak
fp32	23 + 1	256	59 TF
fp16	10 + 1	512	118 TF
bf16	7 + 1	512	118 TF
fp8 e4m3	3 + 1	1024	236 TF
int8	—	1024	236 TOPS

Mixed modes are opportunistic: dynamics never leaves fp32.

c Why a custom ISA

one STENCIL3D retires 7 loads, 7 MACs and a halo exchange that would take 19 generic instructions on a GPU;

no divergence, no speculation, no replay — the worst-case cycle count of a tile is known at compile time (61,900 cycles for the W1 step).